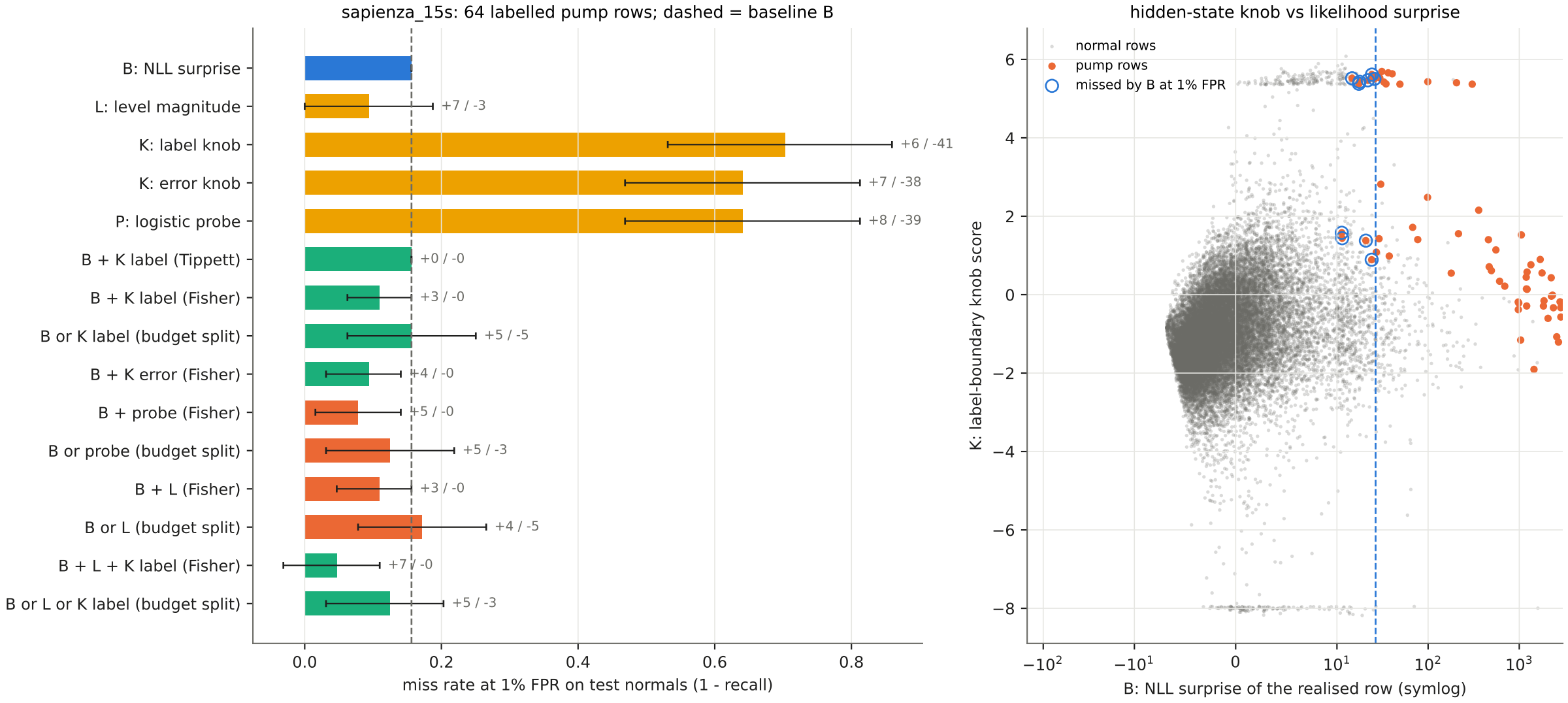

sapienza_15s: 64 labelled pump rows; dashed = baseline B
B: NLL surprise
L: level magnitude
K: label knob
K: error knob
P: logistic probe
B + K label (Tippett)
B + K label (Fisher)
B or K label (budget split)
B + K error (Fisher)
B + probe (Fisher)
B or probe (budget split)
B + L (Fisher)
B or L (budget split)
B + L + K label (Fisher)
B or L or K label (budget split)
+7 / -3
+6 / -41
+7 / -38
+8 / -39
+0 / -0
+3 / -0
+5 / -5
+4 / -0
+5 / -0
+5 / -3
+3 / -0
+4 / -5
+7 / -0
+5 / -3
0.0
0.2
0.4
0.6
0.8
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
pump rows
missed by B at 1% FPR
K: label-boundary knob score
6
4
2
0
−2
−4
−6
−8
−10²
−10¹
0
10¹
10²
10³
B: NLL surprise of the realised row (symlog)